

▼ **Figure 38.17 Non-Bt versus Bt maize.** Field trials reveal that non-Bt maize (left) is heavily damaged by insect feeding and *Fusarium* mold infection, whereas Bt maize (right) suffers little or no damage.



Non-Bt maize

Bt maize

that is toxic to many insect pests (**Figure 38.17**). The Bt toxin used in crops is produced in the plant as a harmless protoxin that only becomes toxic if activated by alkaline conditions, such as in the guts of most insects. Because vertebrates have highly acidic stomachs, protoxin consumed by humans or livestock is rendered harmless by denaturation.

**Biofortification**, improving the nutritional quality of plants, is another strategy in the war on world hunger. For example, about 250,000 to 500,000 children go blind each year because of vitamin A deficiencies. More than half of these children die within a year of becoming blind. In response to this crisis, genetic engineers created “Golden Rice,” a transgenic variety supplemented with transgenes that enable it to produce grain with increased levels of beta-carotene, a precursor of vitamin A. In 2018, after decades of safety testing and legal challenges, Golden Rice received positive food safety evaluations from the governments of Canada, New Zealand, and the United States. There are hopes that the governments of the Philippines and Bangladesh will soon follow suit.

Another major target for biofortification improvement is cassava (*Manihot esculenta*), a starchy root crop that provides carbohydrates for 800 million of the poorest people on our planet (**Figure 38.18**). Researchers are also engineering plants with enhanced resistance to disease. In one case, a transgenic

➔ **Mastering Biology Interview with Dennis Gonsalves: Saving the papaya industry (see the interview before Chapter 35)**



► **Figure 38.18 Engineering the perfect food?** Cassava grows well in poor dry soils but must be supplemented by consumption of other crops. If those crops fail, people suffer malnutrition. What if cassava alone could supply a balanced diet? Transgenic, biofortified cassava plants have been developed with greatly increased levels of iron and beta-carotene (a vitamin A precursor). Other improvements have been made in the size of its storage roots, its ease of processing (less cyanide-producing chemicals), and its resistance to cassava mosaic virus disease.



papaya that is resistant to a ringspot virus was introduced into Hawaii, thereby saving its papaya industry.

Considerable controversy has arisen concerning transgenic crops that are resistant to the herbicide glyphosate. Glyphosate is lethal to a wide variety of plants because it inhibits a key enzyme in a biochemical pathway that is found in plants (and most bacteria) but not in animals. Researchers discovered a bacterial strain that had undergone a mutation in the gene encoding this enzyme that rendered it glyphosate resistant. When this mutated bacterial gene was spliced into the genome of various crops, these crops also became glyphosate resistant. Farmers achieved almost total weed control by spraying glyphosate over their fields of glyphosate-resistant crops. Unfortunately, the overuse of glyphosate created a huge selective pressure on weed species, with the result that many have evolved resistance to glyphosate. There has also been a growing appreciation in recent decades of the role that gut bacteria play in animal and human health, and claims have been made that glyphosate may be having negative effects on the health of humans and livestock by interfering with beneficial gut bacteria.

### Reducing Fossil Fuel Dependency

Global sources of inexpensive fossil fuels, particularly oil, are rapidly being depleted. Moreover, most climatologists attribute global warming mainly to the rampant burning of fossil fuels, such as coal and oil, and the resulting release of the greenhouse gas CO<sub>2</sub>. How can the world meet its energy demands in the 21st century in an economical and nonpolluting way? In certain localities, wind or solar power may become economically viable, but such alternative energy sources are unlikely to fill the global energy demands completely. Many scientists predict that **biofuels**—fuels derived from living biomass—could produce a sizable fraction of the world’s energy needs in the not-too-distant future. **Biomass** is the total mass of organic matter in a group of organisms in a particular habitat. The use of biofuels from plant biomass would reduce the net emission of CO<sub>2</sub>. Whereas burning fossil fuels increases atmospheric CO<sub>2</sub> concentrations, biofuel crops reabsorb by photosynthesis the CO<sub>2</sub> emitted when biofuels are burned, creating a cycle that is carbon neutral.

In working to create biofuel crops from wild precursors, scientists are focusing their domestication efforts on fast-growing plants, such as switchgrass (*Panicum virgatum*) and poplar (*Populus trichocarpa*), that can grow on soil that is too poor for food production. Scientists do not expect the plant biomass to be burned directly. Instead, the polymers in cell walls, such as cellulose and hemicellulose, which constitute the most abundant organic compounds on Earth, would be broken down into sugars by enzymatic reactions. These sugars, in turn, would be fermented into alcohol and distilled to yield biofuels. In addition to increasing plant polysaccharide content and overall biomass, researchers are trying to genetically engineer the cell walls of plants to increase the efficiency of the enzymatic conversion process.